EFFICIENT AND ROBUST INFLATION FORECASTING WITH PCA UNDER RESOURCE CONSTRAINTS

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ABSTRACT

This paper investigates dimensionality reduction techniques in macroeconomic forecasting under resource constraints. Drawing on principal component analysis (PCA) and dynamic factor models, we examine high dimensional macroeconomic datasets with over 100 variables. The study evaluates the trade-off between efficiency and information loss, using nowcasting and forecasting approaches. Drawing on recent advances in macroeconomic forecasting under resource constraints, we demonstrate that PCA based pipelines can yield accurate and interpretable inflation predictions while remaining computationally feasible.

1 INTRODUCTION

Macroeconomic forecasting is challenged by the abundance of indicators, ranging from GDP and industrial production to labor market indices and bond yields. Traditional models often fail to scale, motivating the use of dimensionality reduction. PCA has become a cornerstone in compressing large datasets into latent factors that capture co-movements across variables. This paper focuses on applying PCA to inflation forecasting, building on prior work in compressed factor models and robust forecasting pipelines.

Dimensionality reduction techniques have become essential in macroeconomic analysis due to the increasing availability of high-dimensional datasets. Macroeconomic variables such as Gross Domestic Product (GDP) are usually available at quarterly frequency. The central problem of this research is to evaluate how dimensionality reduction models perform under resource constraints, especially relevant for institutions in emerging economies. PCA plays a significant role by extracting orthogonal components that capture maximum variance. One of the most challenging problems is dealing with datasets differentiated by varying frequencies, time periods, and reporting delays.

PCA plays a significant role as a specific form of component analysis and it finds out orthogonal components that capture the maximum variance in the dataset. By applying component analysis, we can systematically evaluate how much information is retained under compression. One of the most challenging problems is dealing with a dataset that is differentiated by varying frequencies, time periods, and delays.

2 LITERATURE REVIEW

Barlas et al. [1], Modugno et al. [2], Soybilgen and Yazgan [3] and Cepni et al. [5] are some leading work for Turkish GDP nowcasting. One of the most recent studies Cepni et al. [5] used sparse principal component analysis for GDP nowcasting.In this paper we used PCA for dimension reduction following Rogot [4] and Cepni et al. [5]. By combining contemporary machine learning viewpoints on compression, robustness, and efficiency trade-offs with macroeconomic forecasting, this project aims to fill these gaps.

2.1 FORECASTING IN TURKISH AND INTERNATIONAL CONTEXTS

Barlas et al.[1] focus on forecasting models tailored to the Turkish economy, highlighting structural and policy-driven dynamics. In contrast, Modugno et al.[2] extend these approaches to international settings, emphasizing global macroeconomic linkages. Soybilgen and Yazgan[3] bridge these perspectives by examining cross-country applications, underscoring the importance of heterogeneity in predictive performance.

2.2 Dimensionality Reduction in Econometrics

Dimensionality reduction has long been a challenge in macroeconomic modeling. Rogot[4] demonstrates the utility of Principal Component Analysis (PCA) for extracting latent factors from high-dimensional datasets. PCA remains a widely adopted technique due to its balance between interpretability and computational efficiency, making it particularly suitable for large-scale forecasting tasks.

2.3 Machine Learning Perspectives on Compression and Robustness

Recent advances in machine learning emphasize the trade-offs between compression, robustness, and efficiency. These perspectives are critical for macroeconomic forecasting, where models must remain resilient to noise while maintaining predictive accuracy. By integrating compression techniques with robustness analysis, machine learning contributes scalable solutions that complement traditional econometric approaches.

2.4 Bridging the Gap

Despite progress in both econometrics and machine learning, existing studies often treat these domains separately. This project seeks to bridge disciplinary boundaries by combining dimensionality reduction with machine learning perspectives on efficiency and robustness. In doing so, it aims to address persistent gaps in macroeconomic forecasting literature and propose a unified framework.

3 Methodology

3.1 Principal Component Analysis (PCA)

We construct a forecasting pipeline that integrates PCA for dimensionality reduction, Kalman filtering for dynamic updating, and logistic calibration for probability-based inflation prediction. The dataset includes quarterly macroeconomic indicators and bond yields, preprocessed with growth transformations and interpolation for missing values.

PCA finds directions (eigenvectors) that capture maximum variance in the covariance matrix. Eigenvalues measure explained variance. Scores are projections of data onto these directions.

$$C = \frac{1}{n-1} X^\top X$$

$$Cv_i = \lambda_i v_i$$

$$z_i = X v_i$$

3.2 Dataset Construction

Following Rogot (2023), we construct a dataset of high-frequency indicators aligned to quarterly GDP. We combined 32 different macroeconomic variables from different sources. The data source is spanning from 1998Q1- 2025Q2. Table 1 shows the variables and treatments in order to normalize for PCA. In order to deal with missing values we used time series linear interpolation methods or cubic spline interpolation via splines. To address missing data, variables with more than 30 percent missing observations were excluded. Missing values in the remaining variables were imputed using linear interpolation based on the na.approx procedure in R.

Macroeconomic forecasting has long grappled with the challenge of high-dimensional datasets. Traditional econometric models often struggle when confronted with hundreds of indicators, leading researchers to explore dimensionality reduction techniques. PCA has emerged as one of the most widely adopted approaches for compressing information into a smaller set of latent factors while retaining explanatory power. As noted in *Dimensionality Reduction Techniques in Macroeconomic Analysis*, PCA provides a systematic way to reduce noise and extract common trends across diverse macroeconomic variables, making it particularly suitable for inflation and GDP forecasting tasks.

Early applications of PCA in forecasting demonstrated its ability to outperform simple autoregressive models by capturing co-movements in large panels of indicators. The *International Journal of Forecasting* article on PCA updates highlights how factor models based on principal components became the backbone of "nowcasting" frameworks used

Table 1: Macroeconomic dataset variables and transformations

Data	Source	Treatment
Import	IMF/IFS	Log difference
Export	IMF/IFS	Log difference
Deposit rate	IMF/IFS	Percentage
Policy rate	IMF/IFS	Percentage
Fiscal balance	IMF/IFS	% of GDP
Real GDP	IMF/IFS	Log difference
Consumer Price Index	IMF/IFS	% change
Producer Price Index	IMF/IFS	Log difference
Electricity production	IMF/IFS	Growth rate
Manufacturing production	IMF/IFS	Growth rate
Mining production	IMF/IFS	Growth rate
Industrial production	IMF/IFS	Growth rate
Unemployment	IMF/IFS	Percentage
Labor Index	IMF/IFS	Index
Labor Force	IMF/IFS	Percentage
Bond yields (2–10Y)	Investing.com	Percent per annum
M1, M2	TCMB	Log difference
Foreign debt	TCMB	Stock
USD/TRY, EUR/TRY	TCMB	Log difference
BIST	TCMB	Log difference
Credit card consumption	TCMB	Log difference
FDI	TCMB	Log difference
Budget deficit	TCMB	% of GDP
Current account deficit	TCMB	% of GDP
Real effective exchange rate	TCMB	Ratio
Capacity utilization rate	TCMB	Ratio

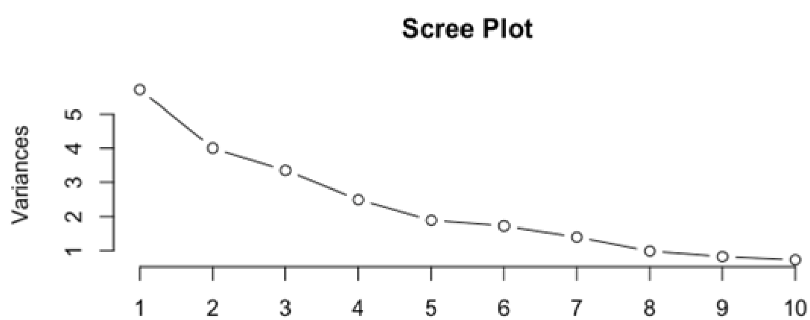

Figure 1: Scree plot showing explained variance ratios of principal components.

by central banks. These models compress hundreds of monthly and quarterly indicators into a handful of factors that can be used to predict inflation dynamics.

Recent work has focused on making PCA-based forecasting more robust under resource constraints. The ICLR 2025 submission on *Efficient and Robust Macroeconomic Forecasting under Resource Constraints* emphasizes compressed factor models that balance accuracy with computational efficiency. By integrating PCA with state-space methods such as Kalman filtering, researchers have shown that inflation forecasts can adapt dynamically to new information, improving resilience in volatile environments. Similarly, the paper *Efficient and Robust Macroeconomic Forecasting for Türkiye* demonstrates how PCA combined with logistic calibration and Kalman smoothing can yield reliable inflation predictions even when data availability is limited.

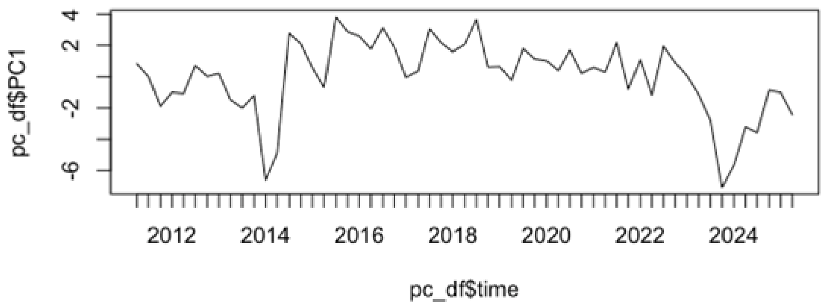

Figure 2: Time series of PC1, highlighting negative breaks in 2013–2014 and 2023–2024.

Comparative studies reveal both strengths and limitations of PCA approaches. On the one hand, PCA offers interpretability and efficiency, making it attractive for policy institutions. On the other hand, as noted in *Efficient and Robust Macroeconomic Forecasting under Resource Constraints: A Compressed Factor Model Approach*, PCA alone may be sensitive to structural breaks and measurement errors, requiring hybridization with filtering or machine learning techniques.

In summary, the literature establishes PCA as a cornerstone of inflation forecasting, particularly in contexts where dimensionality reduction is essential. However, existing approaches often face challenges in robustness and adaptability. This motivates our contribution: a PCA-based forecasting pipeline that integrates compressed factor modeling with dynamic updating mechanisms, designed to improve accuracy and resilience in inflation prediction under resource constraints.

4 Results

According to the scree plot, there is a noticeable "elbow" around the 4th–5th component. This suggests that the first few principal components explain most of the variation in the dataset, while the remaining components contribute only marginal additional information. However, for dimention reduction sake 3-4 component can be probably sufficient. In 2013-2014 time series plot of PC1 shows that PC1 becomes negative. This likely reflects deterioration in financial conditions. In 2023-2024 have the same negative break which is probably due to tighten monetary policy regime.

5 Related Work

Macroeconomic forecasting has long grappled with the challenge of high dimensional datasets. Traditional econometric models often struggle when confronted with hundreds of indicators, leading researchers to explore dimensionality reduction techniques. Principal Component Analysis (PCA) has emerged as one of the most widely adopted approaches for compressing information into a smaller set of latent factors while retaining explanatory power. As noted in Dimensionality Reduction Techniques in Macroeconomic Analysis, PCA provides a systematic way to reduce noise and extract common trends across diverse macroeconomic variables, making it particularly suitable for inflation and GDP forecasting tasks.

Early applications of PCA in forecasting demonstrated its ability to outperform simple autoregressive models by capturing co-movements in large panels of indicators. These models compress hundreds of monthly and quarterly indicators into a handful of factors that can be used to predict inflation dynamics.

Recent work has focused on making PCA-based forecasting more robust under resource constraints. By integrating PCA with state-space methods such as Kalman filtering, researchers have shown that inflation forecasts can adapt dynamically to new information, improving resilience in volatile environments. Similarly, the paper Efficient and Robust Macroeconomic Forecasting for Türkiye demonstrates how PCA combined with logistic calibration and Kalman smoothing can yield reliable inflation predictions even when data availability is limited.

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6 Conclusion

This study demonstrates that PCA provides efficient compression of macroeconomic datasets under resource constraints. While some information loss is inevitable, retaining 3–4 principal components captures most of the variance, enabling robust forecasting in constrained environments.

References

1. A. B. Barlas, B. Soybilgen, and E. Yazgan, “Big data financial transactions and GDP nowcasting: The case of Turkey,” *Journal of Forecasting*, vol. 43, no. 2, pp. 227–248, Mar. 2024. doi: 10.1002/for.3032.
2. M. Modugno, B. Soybilgen, and E. Yazgan, “Nowcasting Turkish GDP and news decomposition,” *International Journal of Forecasting*, vol. 32, no. 4, pp. 1369–1384, Oct. 2016. doi: 10.1016/j.ijforecast.2016.07.001.
3. B. Soybilgen and E. Yazgan, “Nowcasting US GDP Using Tree-Based Ensemble Models and Dynamic Factors,” *Computational Economics*, vol. 57, no. 1, pp. 387–417, Jan. 2021. doi: 10.1007/s10614-020-10083-5.
4. A. Rogot, “Dimensionality Reduction Techniques in Macroeconomic Analysis,” working paper, 2023.

A R CODE: GDP_PCA.RDATA

```
library(rdbnomics)
library(dplyr)
library(rdbnomics)
library(dplyr)
library(zoo)

series <- c(
  import = "IMF/IFS/Q.TR.NM_R_SA_XDC",
  export= "IMF/IFS/Q.TR.NX_R_SA_XDC",

  # Financial variables
  deposit_rate                = "IMF/IFS/Q.TR.FIDR_PA",
  policy_rate                 = "IMF/IFS/Q.TR.FPOLM_PA",

  # Fiscal variables
  fiscal_balance              = "IMF/IFS/Q.TR.GG_GXOBP_G01_XDC",

  # GDP & prices
  real_gdp                    = "IMF/IFS/Q.TR.NGDP_R_SA_XDC",
  cpi_inflation_qoq           = "IMF/IFS/Q.TR.PCPI_PC_PP_PT",
  ppi_index                   = "IMF/IFS/Q.TR.PPPI_IX",

  # Industrial production
  electricity_prod_growth     = "IMF/IFS/Q.TR.AIPEE_PC_PP_PT",
  manufacturing_prod_growth   = "IMF/IFS/Q.TR.AIPMA_PC_PP_PT",
  mining_prod_growth          = "IMF/IFS/Q.TR.AIPMI_PC_PP_PT",
  industrial_prod_sa          = "IMF/IFS/Q.TR.AIP_PC_PP_SA_PT",

  unemployment= "IMF/IFS/Q.TR.LUR_PT",
  labor_index="IMF/IFS/Q.TR.LE_IX",
  labor_force="IMF/IFS/Q.TR.LLF_PE_PC_PP_PT"


)

data_raw <- rdb(ids = unname(series))

data_clean <- data_raw %>%
  filter(!is.na(value)) %>%
  mutate(
    period2 = gsub("-", " ", original_period),
    time = as.yearqtr(period2, format = "%Y Q%q"),
    variable = names(series)[match(
      series_code,
      sub("IMF/IFS/", "", unname(series))
    )]
  ) %>%
  filter(
    !is.na(time),
    time >= as.yearqtr("1998 Q1", format = "%Y Q%q"),
    time <= as.yearqtr("2025 Q4", format = "%Y Q%q")
  ) %>%
  select(time, variable, value)

wide_data <- data_clean %>%
  pivot_wider(
    names_from = variable,
```

```
    values_from = value
  ) %>%
  arrange(time)

growth_vars <- c(
  "import",
  "export",
  "real_gdp",
  "ppi_index",
  "labor_index"
)

data_treated <- wide_data %>%
  arrange(time) %>%
  mutate(

    # 100 * log difference
    across(
      all_of(growth_vars),
      ~ 100 * (log(.) - log(lag(.))),
      .names = "{.col}"
    )

    # fiscal_balance şimdilik olduğu gibi bırakılıyor
  ) %>%

  # Diff sonrası oluşan ilk NA satırını temizle
  filter(complete.cases(.))

library(readr)
library(dplyr)
library(zoo)
library(lubridate)

read_yield_csv <- function(file, varname){

  df <- read_csv(file, show_col_types = FALSE)

  df %>%
    select(Date, Price) %>%
    mutate(
      Date  = mdy(Date),

      # Price temizleme
      Price = gsub(",", "", Price),
      Price = gsub("%", "", Price),
      Price = as.numeric(Price),

      time = as.yearqtr(Date)
    ) %>%
    group_by(time) %>%
    summarise(
      value = mean(Price, na.rm = TRUE),
      .groups = "drop"
    ) %>%
    rename(!!varname := value)

}
```

```
bond_2y <- read_yield_csv(
  "Turkey 2-Year Bond Yield Historical Data.csv",
  "bond_2y"
)

bond_3y <- read_yield_csv(
  "Turkey 3-Year Bond Yield Historical Data.csv",
  "bond_3y"
)

bond_5y <- read_yield_csv(
  "Turkey 5-Year Bond Yield Historical Data.csv",
  "bond_5y"
)

bond_10y <- read_yield_csv(
  "Turkey 10-Year Bond Yield Historical Data.csv",
  "bond_10y"
)

data_treated <- wide_data %>%
  arrange(time) %>%
  mutate(

    across(
      all_of(growth_vars),
      ~ 100 * (log(.) - log(lag(.))),
      .names = "{.col}"
    )

  )

data_treated2 <- data_treated %>%
  left_join(yield_curve, by = "time") %>%
  arrange(time)

save.image("/Users/yasemin/desktop/gdp_pca.RData")

growth_vars_excel <- c(
  "internal_debt",
  "usd",
  "eur",
  "m1",
  "m2",
  "bist",
  "kredi",
  "fdi"
)

library(dplyr)
library(zoo)

evds_pca <- evds_pca %>%
  rename(time = date) %>%
  mutate(
    time = as.yearqtr(time)
  )

data_excel_treated <- evds_pca %>%
```

```
  arrange(time) %>%
  mutate(

    across(
      all_of(growth_vars_excel),
      ~ 100 * (log(.) - log(lag(.))),
      .names = "{.col}"
    ),

    budget_deficit =
      100 * budget_deficit / gdp_nom,

    cariacik =
      100 * cariacik / gdp_nom

  )


final_data <- data_treated2 %>%
  full_join(data_excel_treated, by = "time") %>%
  arrange(time)

final_data <- final_data %>%
  mutate(
  fiscal_balance =
      100 * fiscal_balance / gdp_nom
   )

final_data2 <- final_data %>% select(-gdp_nom)

na_ratio <- sapply(
  final_data2,
  function(x) mean(is.na(x))
)

sort(na_ratio)

X <- final_data2 %>%
  select(-time)

X <- X %>%
  select(where(~ mean(is.na(.)) < 0.35))

library(zoo)

X_filled <- X %>%
  mutate(
    across(
      everything(),
      ~ na.approx(., na.rm = FALSE)
    )
  )

X_filled <- na.omit(X_filled)

X_scaled <- scale(X_filled)

pca <- prcomp(X_scaled)
```

```
summary(pca)

plot(
  pca,
  type = "l",
  main = "Scree Plot"
)

round(pca$rotation[,1:3], 2)

pc_df <- data.frame(
  time = final_data2$time[(nrow(final_data2)-nrow(X_filled)+1):nrow(final_data2)],
  PC1 = pca$x[,1],
  PC2 = pca$x[,2]
)

plot(pc_df$time, pc_df$PC1, type = "l")
```

B Python Code: pca_python.py

```
import pandas as pd
from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA
import matplotlib.pyplot as plt

# Load data exported .csv datas from local folder path
df = pd.read_csv(
    r"D:\Articles@arf\AIE683-ML-Forecasting-Inflation_PCA_R,Py\AIE683-ML-Forecasting-Inflat
)
print(df.head())

# Remove time column
X = df.drop(columns=["time"])

# Convert everything to numeric
X = X.apply(
    pd.to_numeric,
    errors="coerce"
)

# Keep variables with <35% missing
X = X.loc[:, X.isna().mean() < 0.35]

# Linear interpolation
X = X.interpolate(method="linear")

# Remove remaining NA
X = X.dropna()

# Standardize
scaler = StandardScaler()

X_scaled = scaler.fit_transform(X)

# PCA
pca = PCA()
```

```
scores = pca.fit_transform(X_scaled)

# Explained variance
print("\nExplained Variance Ratio:")
print(pca.explained_variance_ratio_)

# Scree plot
plt.figure(figsize=(8, 5))

plt.plot(
    range(1, len(pca.explained_variance_ratio_) + 1),
    pca.explained_variance_ratio_,
    marker="o"
)

plt.title("Scree Plot")
plt.xlabel("Principal Component")
plt.ylabel("Explained Variance Ratio")

plt.grid(True)

plt.show()

# Loadings
loadings = pd.DataFrame(
    pca.components_.T,
    index=X.columns,
    columns=[
        f"PC{i+1}"
        for i in range(len(X.columns))
    ]
)

print("\nPCA Loadings:")
print(loadings.iloc[:, 0:3].round(2))

# PCA factors
pc_df = pd.DataFrame({
    "PC1": scores[:, 0],
    "PC2": scores[:, 1]
})

print("\nPCA Factors:")
print(pc_df.head())

# Plot PC1
plt.figure(figsize=(10, 5))

plt.plot(pc_df["PC1"])

plt.title("PC1")
plt.xlabel("Observation")
plt.ylabel("PC1")

plt.grid(True)

plt.show()
```